

MODULE 1

Data Access, Cleaning & Analysis

Excel Online | Getting data you can trust

Day 1 • 09:00 – 11:45

WHERE THIS MODULE SITS IN THE DATA LIFECYCLE



FIVE KEY HABITS FROM THIS MODULE

- Audit before you act — scan every new dataset for quality issues before building anything on top of it.
- Clean first, analyse second — the quality of your analysis is bounded by the quality of your data.
- Convert data to a Table — structured Tables make formulas, filters, and PivotTables more reliable.
- Use XLOOKUP, not VLOOKUP — XLOOKUP is more robust, more readable, and won't silently break.
- Choose the right summary tool — formulas for fixed structures, PivotTables for exploration.

FIRST-LOOK AUDIT CHECKLIST

Duplicates	Are there repeated rows that should appear only once?
Blank cells	Are there gaps in key columns? Are there gaps in the data or gaps in recording?
Number formats	Are numbers stored as text? Check: left-aligned numbers are text, not numeric.
Date formats	Are dates consistent across the column? Mixed formats break sorting and formulas.
Inconsistent text	Are category values entered consistently? 'Full Time', 'full time', 'FT' are three different values.
Trailing spaces	Are there invisible spaces in cells? Use =TRIM() to detect and fix them.

CORE FORMULAS REFERENCE

COUNTIF

=COUNTIF(range, criteria)

Count cells in a range that meet one condition. E.g. count all 'Not Complete' in column C.

COUNTIFS

=COUNTIFS(range1, criteria1, range2, criteria2)

Count cells meeting multiple conditions simultaneously.

SUMIF

=SUMIF(range, criteria, sum_range)

Sum values in one column where a corresponding column meets a condition.

SUMIFS

=SUMIFS(sum_range, range1, criteria1, range2, criteria2)

Sum values meeting multiple conditions across multiple columns.

IF

=IF(logical_test, value_if_true, value_if_false)

Return different values based on a condition. Useful for flagging compliant/non-compliant records.

IFERROR

=IFERROR(formula, value_if_error)

Return a clean value (e.g. 0 or blank) instead of showing an error. Wrap any formula shared with stakeholders.

VLOOKUP vs XLOOKUP — KNOW THE DIFFERENCE

VLOOKUP	XLOOKUP
=VLOOKUP(value, table, col_num, FALSE)	=XLOOKUP(value, lookup_array, return_array, "not found")
• Can only look right — lookup column must be leftmost • References columns by number — breaks if columns are inserted • No built-in 'not found' handling without IFERROR • Still common in legacy files — know how to read it	• Looks in any direction — left, right, or across rows • References arrays directly — survives structural changes • Built-in 'if not found' argument — no IFERROR needed • Use this for all new formulas in Excel Online

PIVOTTABLE QUICK GUIDE

When to use	When you want to explore how a metric breaks down by different categories — and may want to change that breakdown without rewriting formulas.
When not to use	When you need a fixed, portable summary others can read and audit without pivot experience. Use COUNTIF/SUMIF instead.
Build one	Click anywhere in your Table → Insert → PivotTable → choose rows, columns, and value field.
Refresh it	Right-click the PivotTable → Refresh. Always refresh before sharing a report built on a pivot.
Key tip	Sort your row labels by value descending so the most significant category appears at the top.

Remember

A formula that returns a plausible number is not the same as a formula that returns the correct number. After building any summary, check two or three values manually against the raw data before sharing the output.

